Do Now · retrieval

Programming Techniques · Scope and Lifetime

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does return do in a method?

Q2

What is a condition-controlled loop?

Q3

Why might binary search outperform linear search?

Q4

Simplify ¬¬B.

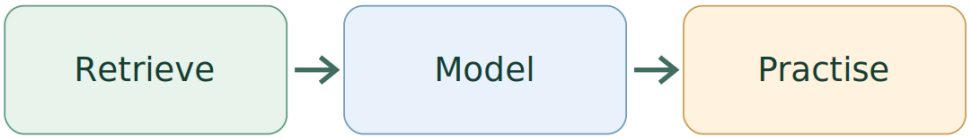
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Explain the important decision — The class field and method-local variable are different storage locations. C# allows a local to hide a field; this does not permit redeclaring any local in a nested block.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

scope — Where a name is accessible

local — Declared in a limited scope

class field — Variable belonging to a class/object

shadowing — Inner name hides an outer name

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Spot two variables with one name

Follow the worked method

1. Identify both variables — Program.score is a class-level static field, initially 10. Main declares its own local score, initially 3.

2. Resolve each name — The unqualified score inside Main refers to the local. Program.score explicitly selects the class field; the field has not changed.

3. Explain the output and scope — The output is 3 then 10. Another method cannot directly access Main's local score. Restricting scope reduces unintended changes to shared state.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Discuss, then respond
A parameter n is set to 4. What is visible inside the function?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Explain one benefit of local variables. [2]

Question 2

Apply independently

A method declares int total=0. Can main read total directly? [1]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Define variable shadowing. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: every local variable is visible throughout the program.

Use key vocabulary from memory

1. Explain scope in your own words.

2. Explain local in your own words.

3. Explain class field in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.